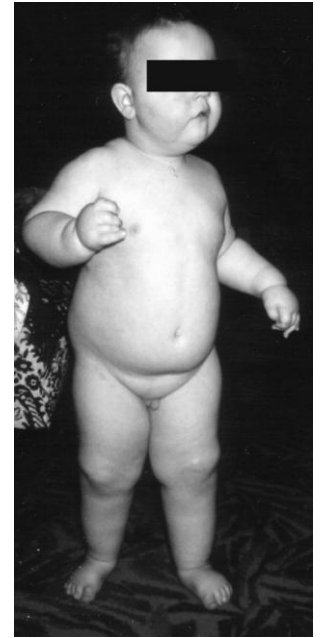


FTM IMCQ4

Biochemistry and Genetics

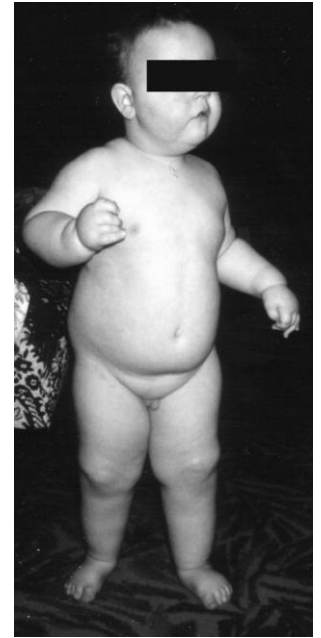
KEY File

A 2-year-old boy is brought to the physician by his parents for evaluation of developmental delay and poor muscle tone. His parents report that he has an insatiable appetite and has begun gaining excessive weight. Physical examination shows hypotonia, almond shaped eyes and small hands and feet. Genetic testing confirms the diagnosis of Prader-Willi syndrome. Which of the following best explains the genetic mechanism underlying this patient's condition?



- A. Deletion of maternal SNRPN gene
- B. Paternal uniparental disomy of chromosome 15
- C. Mutation in the maternal UBE3A gene
- D. Maternal uniparental disomy of chromosome 15
- E. Mutation in the paternal UBE3A gene

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SOM.MKIII.BPM1.1.FTM.3.GNET.0064
Identify imprinting defects (Prader Willi syndrome, Angelman syndrome)
SOM.MKIII.BPM1.1.FTM.3.GNET.0065
Explain the phenomenon of uniparental disomy and microdeletion in Prader-Willi and Angelman syndrome

A 38-year-old man is brought to the physician by his wife because of a two-month history of involuntary movements, mood changes, and difficulty concentrating. His father died in his early 50s after developing similar symptoms. Neurologic examination shows choreiform movements and mild cognitive impairment. Genetic testing confirms an expanded CAG trinucleotide repeat in the HTT gene. Which of the following best explains the earlier onset and increased severity of symptoms in this patient compared to his father?

- A. Imprinting
- B. Digenic inheritance
- C. Variable expressivity
- D. Anticipation
- E. Heteroplasmy

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- E. Heteroplasmy

A 12-year-old girl is brought to the physician by her parents because of a 1-year history of progressive muscle weakness and exercise intolerance with recent episodes of vomiting. Her mother reports that several maternal relatives have similar symptoms, though the severity varies widely. Laboratory studies show elevated serum lactate levels. A muscle biopsy specimen shows ragged red fibers. Genetic testing confirms a mitochondrial DNA mutation causing MELAS. Which of the following best explains the variability in symptom severity among affected family members?

- A. Mitochondrial heteroplasmy
- B. Triplet repeat expansion within a mitochondrial gene
- C. Genomic imprinting of mitochondrial genes
- D. Increased physical activity
- E. Epigenetic silencing of the paternal mitochondria

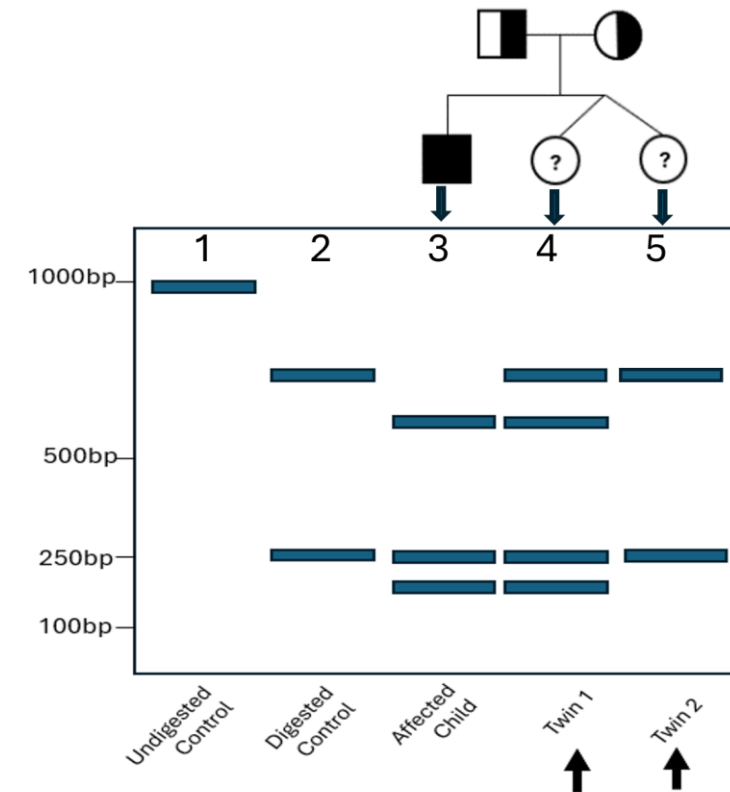
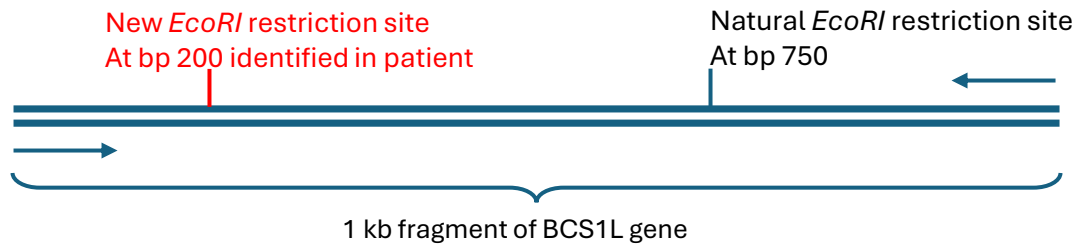
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SOM.MKIII.BPM1.1.FTM.3.GNET.0062

Identify the phenomenon of heteroplasmy in mitochondrial disorders and indicate its genetic basis

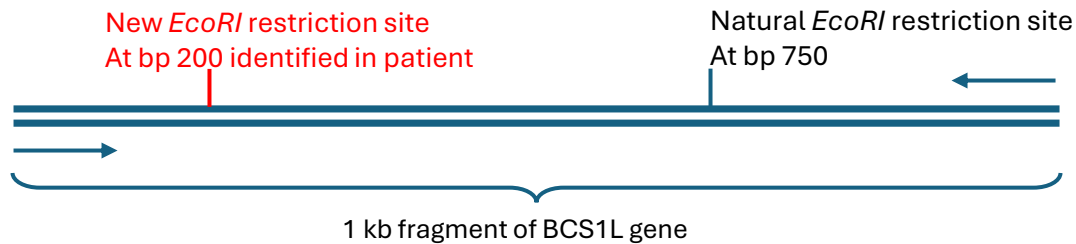
- ✓ A. Mitochondrial heteroplasmy
- B. Triplet repeat expansion within a mitochondrial gene
- C. Genomic imprinting of mitochondrial genes
- D. Increased physical activity
- E. Epigenetic silencing of the paternal mitochondria

A 2-year-old boy is brought to the physician by his mother because of a 1-year history of growth delay. Laboratory studies show hepatic iron overload, aminoaciduria, cholestasis, and lactic acidosis. A diagnosis of GRACILE syndrome is made. Exome sequencing of the patient and his parents (trio) shows the patient is homozygous for c.330G>A variant in the *BCS1L* gene, and both parents are carriers. The c.330G>A variant introduces a new *EcoRI* restriction site into the coding region of the gene. The parents of the patient have newborn fraternal twin girls. An RFLP test is developed for the family, and the results are shown in the figure. Which of the following statements is most accurate for the twin girls in lanes 4 and 5 following RFLP analysis?

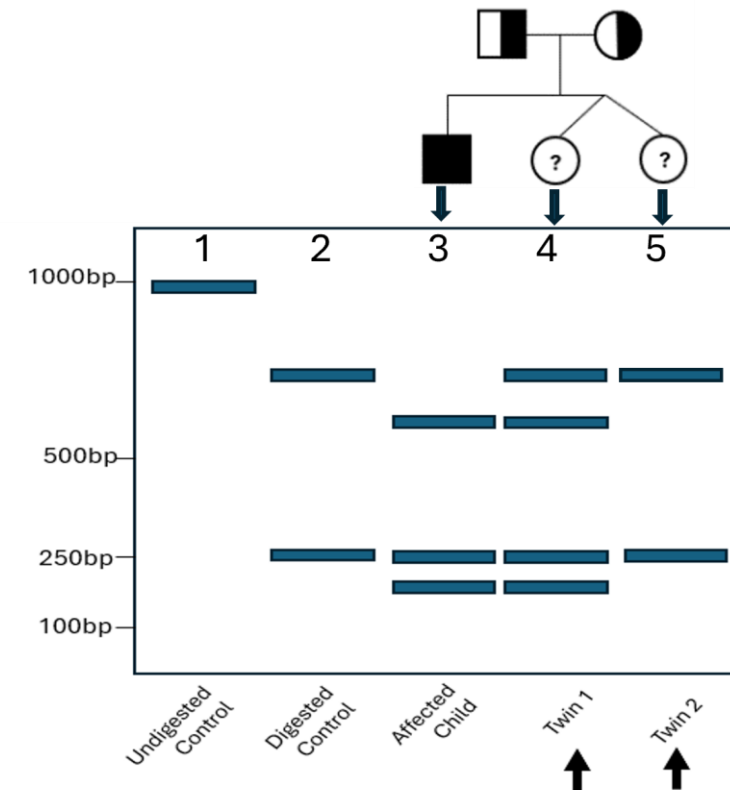


- A. Twin 1 is affected, and Twin 2 is healthy
- B. Both twins are healthy
- C. Twin 1 is a carrier and Twin 2 is healthy
- D. Twin 1 is a carrier and Twin 2 is affected
- E. Both twins are carriers

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- B. Both twins are healthy
- ✓ C. Twin 1 is a carrier and Twin 2 is healthy
- D. Twin 1 is a carrier and Twin 2 is affected
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A 19-year-old man comes to the physician seeking advice because he has a positive family history of an autosomal recessive disorder. The genetic variation in his family is well documented in the literature, and an ASO test is available. This test is offered to the patient to determine his carrier status. Which of the following types of genetic variations is this test most likely to identify?

- A. An extra chromosome
- B. 500 Kb deletion
- C. A point mutation
- D. 5 Mb deletion
- E. A missing chromosome
- F. Chromosome translocation

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SOM.MK.III.BPM1.1.FTM.3.GNET.0203 Explain the concept of hybridization: between a complementary single stranded DNA probe and its target (DNA or RNA); between an antibody and its protein target

- A. An extra chromosome
- B. 500 Kb deletion
- ✓ C. A point mutation
- D. 5 Mb deletion
- E. A missing chromosome
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A 15-month-old boy is brought to the physician by his parents because of global developmental delay, and lack of language attainment. His symptoms do not match a known syndrome. Genetic tests are performed. Microarray CGH analysis is normal, but exome sequencing shows a pathogenic variant in a gene known to play a role in a rare form of autism. Which of the following genetic aberrations is most likely observed in the DNA of this patient?

- A. Copy number variant
- B. Triplet repeat expansion
- C. One base pair deletion
- D. A change in the centromere
- E. A change in the telomere

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A 5-year-old girl is brought to the physician by her mother for a follow-up examination for mild learning difficulties and growth delays. Her mother reports that her daughter has strong socialization skills but has weak verbal communication. Physical examination shows the findings in the photograph. Genetic testing shows that she has Wolf-Hirschhorn syndrome due to a 1.9 Mb deletion from chromosome 4p16.3 to the terminus of the chromosome. Which of the following tests was most likely used to obtain this diagnosis?

- A. SKY-FISH
- B. Western blot
- C. Enzyme activity
- D. G-banding
- E. Array CGH



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SOM.MKIII.BPM1.1.FTM.3.GNET.0074: Describe the strategy behind the microarray: CGH Microarray, SNP Microarray, cDNA Microarray

- A. SKY-FISH
- B. Western blot
- C. Enzyme activity
- D. G-banding
- ✓ E. Array CGH



A 5-year-old boy is brought to the physician by his parents for evaluation of developmental delay and behavioral issues. Physical examination shows a long face, large ears, and macroorchidism. Family history reveals intellectual disability in several male relatives on the maternal side. Fragile X syndrome is suspected and a genetic test that can detect CGG trinucleotide repeat expansions in the *FMR1* gene is ordered. Which of the following is the most appropriate diagnostic test to confirm the suspected condition?

- A. Incorporation of a ddNTP
- B. Hybridization of a fluorescent probe
- C. Formation of a cDNA
- D. Polymerase chain reaction
- E. Identification of an SNP
- F. Staining with Giemsa



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A 2-year-old girl is brought to the clinic by her parents because of global developmental delay. Physical examination shows the patient is below the 10th percentile for height and weight, has mildly dysmorphic facial features, and has limited speech. Neurological examination shows hypotonia and mild motor incoordination. Standard karyotyping does not show any chromosomal abnormalities. Microarray CGH analysis is performed. Which of the following would most likely be detected by this test?

- A. Point mutations
- B. Frameshift mutation
- C. Copy number variations
- D. Chromosome inversions
- E. Triplet repeat expansions

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FTM IMCQ4

BCHM=7

A 22-year-old college student comes to the emergency department because of palpitations, anxiety, and tremulousness after taking several over-the-counter caffeine tablets to stay awake for exams. Physical examination shows tachycardia and fine hand tremor. Results of laboratory studies are within the reference ranges. Which of the following intracellular changes most likely explains this patient's symptoms?

- A. Activation of adenylate cyclase
- B. Decreased hydrolysis of cAMP to 5'-AMP
- C. Inhibition of guanylate cyclase
- D. Activation of protein kinase C via IP3/DAG pathway
- E. Opening of GABA-A chloride channels

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A 50-year-old man comes to the physician for a routine examination. He has a family history of type-2 diabetes mellitus and hypertension. His blood pressure is 135/85 mm Hg. Blood samples are obtained for estimation of hemoglobin A1c (HbA1c). Which of the following best describes the hemoglobin estimated in this patient?

- A. Contains four alpha-globin chains
- B. Contains four beta-globin chains
- C. Formed by phosphorylation of serine residues
- D. Formed by non-enzymatic addition of glucose
- E. Contains only beta-pleated sheets
- F. Contains valine at the 6th position of beta-globin

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A 72-year-old woman is brought to the physician by her daughter because of a 2-year history of progressive memory loss and difficulty performing daily tasks. Her daughter reports that she often forgets recent conversations, misplaces objects, and has trouble managing her finances. Physical examination shows no motor weakness or sensory deficits. Mini-Mental State Examination score is 19/30. Laboratory studies, including thyroid function tests and vitamin B12 levels, are within the reference ranges. MRI of the brain shows generalized cortical atrophy, most pronounced in the hippocampus. Which of the following is the most likely underlying mechanism of this patient's condition?

- A. Non-enzymatic glycation of collagen
- B. Premature death of sickled RBCs
- C. Abnormal CFTR protein folding
- D. Abnormal proteolysis of amyloid precursor protein
- E. Nondisjunction of chromosome 21 during meiosis I

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A 55-year-old man is brought to the emergency department by his daughter because of a severe, anterior chest pain of sudden onset lasting 30 minutes. He also has shortness of breath and sweating at rest. He has hypertension, hypercholesterolemia, and type 2 diabetes mellitus. His pulse is 110/min, respirations are 20/min, and blood pressure is 150/85 mm Hg. Cardiac examination shows a grade 2/6 systolic murmur at the apex which radiates to the axillae. A 12-lead electrocardiogram shows ST segment elevation in leads II, III, and aVF. He is given 325 mg chewable aspirin to prevent further thrombosis, and sublingual nitrate as a symptomatic relief to his chest pain. Which of the following best explains the rationale behind the administration of the drug to prevent further thrombosis?

- A. Increased thromboxane production by platelets
- B. Increased prostacyclin production by platelets
- C. Increased thromboxane production by endothelial cells
- D. Increased prostacyclin production by endothelial cells
- E. Decreased thromboxane production by platelets
- F. Decreased prostacyclin production by platelets
- G. Decreased thromboxane production by endothelial cells
- H. Decreased prostacyclin production by endothelial cells

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A 1-year-old girl is brought to the emergency department by her parents because of severe pain and swelling of her fingers and toes. Physical examination shows conjunctival pallor, scleral icterus (yellowing of the whites of her eyes) and dactylitis (swellings in the fingers and toes). Hemoglobin genotyping shows she has HbSS. Which of the following changes is most likely in the affected protein in this patient?

- A. Substitution of sulfur containing amino acid with proline
- B. Substitution of a basic amino acid with an acidic amino acid
- C. Substitution of negatively charged amino acid with hydrophobic amino acid
- D. Becomes more negatively charged than HbA
- E. Substitution of a branched chain amino acid with positively charged amino acid
- F. Alpha helices are converted to beta-pleated sheets

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SOM.MK.I.BPM1.1.FTM.3.BCHM.0090; Relate that missense mutations can alter the solubility, function, charge and electrophoretic properties of proteins.

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- E. Substitution of a branched chain amino acid with positively charged amino acid
- F. Alpha helices are converted to beta-pleated sheets

A 53-year-old man comes to the physician because of a 3-month history of hair thinning, weight gain and acne. Physical examination shows moon-shaped facial features, truncal obesity, red-purple abdominal striae and thin extremities. Laboratory studies show serum cortisol level is 465 $\mu\text{g/dL}$ (normal range = 5 – 23 $\mu\text{g/dL}$). Which of the following receptor types most likely mediates the primary cellular effects of the excess hormone causing this patient's findings?

- A. Tyrosine kinase
- B. Gated-ion channel
- C. Serine/threonine kinase
- D. Cell-surface
- E. G-protein coupled
- F. Intracellular

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A 3-year-old boy is brought to the physician by his mother because of a 1-week history of fever and productive cough. His mother says that his skin is salty, and his stools are bulky and greasy. Genetic studies show homozygosity for the most common mutation for the disorder. Which of the following mutations is most likely in this patient?

- A. Missense mutation
- B. Nonsense mutation
- C. Promoter mutation
- D. Deletion of 3 base pairs
- E. Splice site mutation

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